

Quality Management

6 — The seven statistical tools of quality

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Today

From opinions to evidence

ISO 9000 lists **evidence-based decision making** among its seven quality management principles. Today is the toolbox that makes it possible.

Seven simple tools — eight, as we will see — each answering **one question** about a process. For every tool: **an everyday example**, then **a real quality investigation** that we follow from the first symptom to the final fix.

1

Why statistics

variation, and why quality is about managing it

2

See and count

flowchart, check sheet, Pareto chart

3

Find the causes

cause-and-effect, stratification, scatter

4

Describe variation

samples, mean and σ , histogram

5

Control variation

control charts, process capability

PART 1

Why statistics?

No two things a process makes are ever identical

Every process varies

The one fact this whole lecture is built on

In your day

- Your commute is never exactly 25 minutes.
- Two espressos from the same bar, same barista, never taste the same.
- A “500 g” packet of pasta weighs 497 g, or 506 g.
- Your phone battery lasts a different time every day.

In a factory

- Two shafts turned one after the other on the same lathe differ by a few micrometres.
- Two toasters from the same line: one works for ten years, one fails in a week.
- Fill weight, torque, temperature, solder quality: everything varies.

Variation is the enemy of quality

A customer never experiences your *average*. They experience the one toaster they bought. Quality management is, in large part, **the management of variation**, and statistics is the language in which variation is described.

Two questions run through the whole lecture: **how much** does it vary? And **why**?

Why use statistical tools at all?

Six reasons, all of them practical

- **Facts, not opinions.** “The line feels slow” starts an argument. A chart of cycle times ends one — and the decision survives when its author leaves the room.
- **Causes, not symptoms.** Sorting out bad parts treats the symptom. The tools trace the effect back to the cause, so the problem does not return.
- **Priorities.** Money and people are always limited. The tools show which few problems are worth attacking first.
- **Signal versus noise.** Not every bad day is a problem. Reacting to normal variation makes a process *worse*; ignoring a real change lets it drift.
- **A common language.** A chart on the wall is read the same way by the operator, the engineer and the plant manager.
- **Required, not optional.** ISO 9001 §9.1: decide what to monitor and measure, and how to analyse and evaluate it. §10.2: determine the causes of every nonconformity.

Seven tools, and a promise

Kaoru Ishikawa and the Japanese quality circles

In the 1960s Ishikawa (1915–1989) was building **quality circles**: small groups of shop-floor workers solving their own problems. They needed tools anyone could use *with a pencil* — no statistician required.

“As much as **95 % of quality problems** can be solved with **seven fundamental tools**.”

— K. Ishikawa

His point: do not reach for advanced statistics first. Built on Shewhart (control chart, 1924), Pareto and Juran (the vital few) and Deming’s lectures in Japan (1950) — the pioneers of quality management.

Which seven? Two versions of the list are in use:

Tool	Ishikawa’s list	Many later lists
Check sheet	✓	✓
Pareto chart	✓	✓
Cause-and-effect diagram	✓	✓
Histogram	✓	✓
Scatter diagram	✓	✓
Control chart	✓	✓
Stratification	✓	—
Flowchart	—	✓

We cover **both**: the flowchart is where every investigation starts, stratification is where many are solved.

One question per tool

If you remember one slide from today, make it this one

1 • Flowchart

What actually happens, step by step?

2 • Check sheet

How often does each thing happen — when and where?

3 • Pareto chart

Which few problems cause most of the damage?

4 • Cause and effect

What could be causing this effect?

5 • Stratification

Is it the same for every machine, shift, supplier, day?

6 • Scatter diagram

Do two variables move together?

7 • Histogram

What does the variation look like?

8 • Control chart

Is the process stable over time — has something changed?

And one more question, at the very end

Process capability: once the process is stable, can it actually meet the specification?

Where the tools fit

PDCA and Six Sigma's DMAIC use the same toolbox

PDCA	DMAIC	The question	Tools
Plan	Define	What is the problem, and where in the process?	flowchart, Pareto chart
	Measure	How big is it, how often, how variable?	check sheet, histogram, capability
	Analyse	Why does it happen?	cause and effect, stratification, scatter
Do	Improve	Does the fix work?	Pareto before / after, scatter, histogram
Check · Act	Control	Will it stay fixed?	control chart, capability

DMAIC IN ONE LINE

Define, Measure, Analyse, Improve, Control: the improvement cycle of **Six Sigma** (Motorola, 1980s). It is PDCA with the measurement steps written out — and the seven tools do most of the work in it.

The case we follow today

A toaster factory, and one bad week

A plant assembles toasters. In one week, final testing rejects **145 units**. Management sets a target: **reduce defects by 25 %**.

Nobody knows yet what is wrong. Opinions are plentiful:

- “It’s the new board supplier.”
- “It’s the weekend crew.”
- “Final testing has become too strict.”

We follow the investigation tool by tool. Each tool takes one step, and each step starts from the output of the one before.

The path

- 1 flowchart → where are defects found?
- 2 check sheet → which defects, which days?
- 3 Pareto → which defect first?
- 4 cause and effect → what could cause it?
- 5 stratification → is it the same every day?
- 6 scatter → is the suspect guilty?
- 7 control chart → did the fix work, and does it hold?

BEFORE WE START

You are the quality manager. What would you look at first? Write it down — we compare at the end.

PART 2

See the process, count the problems

Flowchart · Check sheet · Pareto chart

Flowchart

Tool 1 · What actually happens, step by step?

A **visual map of the sequence of a process**: the flow of information, tasks, people, material or decisions.

- **Makes the complex simple.** One page instead of a forty-page procedure.
- **Creates a common understanding.** Everyone sees the same process — often for the first time.
- **Shows where quality is decided.** Every decision diamond is a point where a defect is caught — or missed.
- **Built by the people who do the work.** Sticky notes on a wall, one step each, then put in order. The team knows things no manager does.

Four versions of the same process

1. what management *thinks* it is
2. what the written procedure (SOP) *says*
3. what *actually happens* on the floor
4. what it *could be*

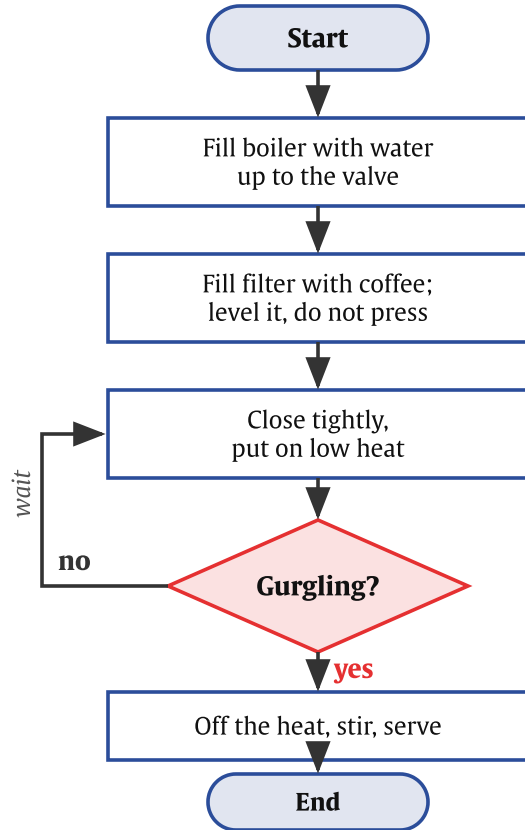
Draw number 3 first. The gap between 2 and 3 is a quality finding in itself.

“If you can’t describe what you are doing as a process, you don’t know what you’re doing.”
— W. Edwards Deming

Flowchart — two examples

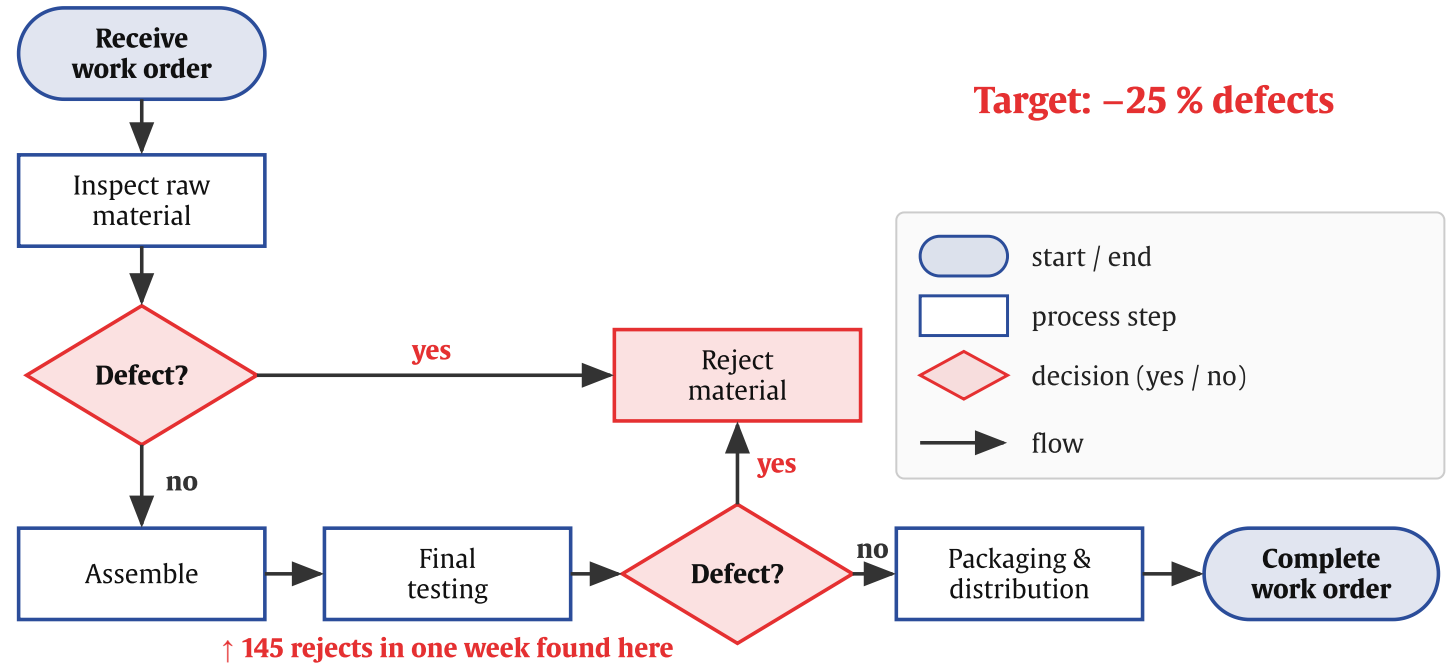
A moka pot, and the toaster line

EVERYDAY · COFFEE WITH A MOKA



Quality is decided at the diamond: leave it on the heat too long and the coffee burns.

QUALITY INVESTIGATION · TOASTER ASSEMBLY



Two inspection points. The rejects are found at **final testing** — so that is where the data will be collected.

Check sheet

Tool 2 · How often does each thing happen — when and where?

A form **prepared before** collecting, so data are recorded where and when they occur — already organised for analysis. The simplest tool and the most important: every other tool needs data, and bad data cannot be rescued by good statistics.

- **Tally sheet** — counts by category: defect type × day.
- **Location sheet** — a drawing of the part; mark where each defect is. Scratches all on one corner point at the fixture.
- **Distribution sheet** — tally measurements into ranges: a histogram that draws itself.
- **Checklist** — was every step done? A pilot’s pre-flight check.

Metadata: data about the data

Who recorded it · **when** (date, shift, time) · **where** (line, machine). Skip it and you can never stratify later.

EVERYDAY · “WHY WAS I LATE?” — ONE SEMESTER

Reason	Mon	Tue	Wed	Thu	Fri	Total
Bus late or missed						14
Overslept						7
Looking for parking						4
Coffee queue						3
Other						2
Total	8	5	6	5	6	30

Recording *when* pays off at once: oversleeping is a **Monday** problem (4 of 7).

Check sheet — the toaster line

One week of final-testing rejects, by defect type and day

Recorded by: final-test inspector · Recorded on: week 40 · Product: 2-slice toaster · Process: final testing · Equipment: test bench 2

Defect	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Control PCB	6	4	1	10	17	10	15	63
Heater element	2	0	6	6	2	0	7	23
Solenoid	4	4	5	1	4	0	5	23
Control knob	1	3	0	0	3	2	2	11
Latch	0	0	2	3	0	2	2	9
Lever	2	2	1	1	0	1	0	7
Thermostat	1	1	0	1	1	1	0	5
Chassis damage	0	2	1	0	1	0	0	4
Total	16	16	16	22	28	16	31	145

What the target means

145 rejects a week $\approx$ 20.7 a day.
 –25 % = **36 fewer a week**, about **5 fewer a day**.

LOOK BEFORE READING ON

Which defect dominates? Now read the *Control PCB* row day by day: anything odd?

The first answer gives the Pareto chart. The second comes back in Part 3.

Pareto chart

Tool 3 · Which few problems cause most of the damage?

A bar chart of categories **sorted from largest to smallest**, with a **cumulative-percentage** line.

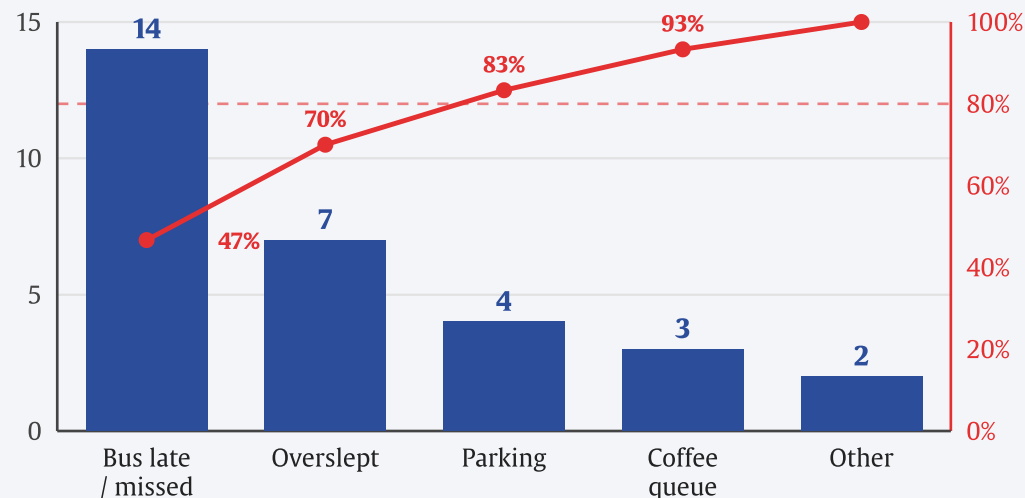
Vilfredo Pareto, Italian economist (1896): about 80 % of the land in Italy was owned by about 20 % of the people. **Joseph Juran** carried the idea into quality: the *vital few* and the *trivial many* — later the “useful many”, because small problems still matter, just not first. 80/20 is a rule of thumb, not a law.

Building one

1. Choose categories and a measure: count, cost or time.
2. Sort descending; “Other” always goes last.
3. Compute % and cumulative %.
4. Bars on the left axis, cumulative line on the right.

Excel: *Insert* → *Statistic chart* → *Pareto*, or sort and build a combo chart by hand.

EVERYDAY · THE 30 LATE ARRIVALS, SORTED

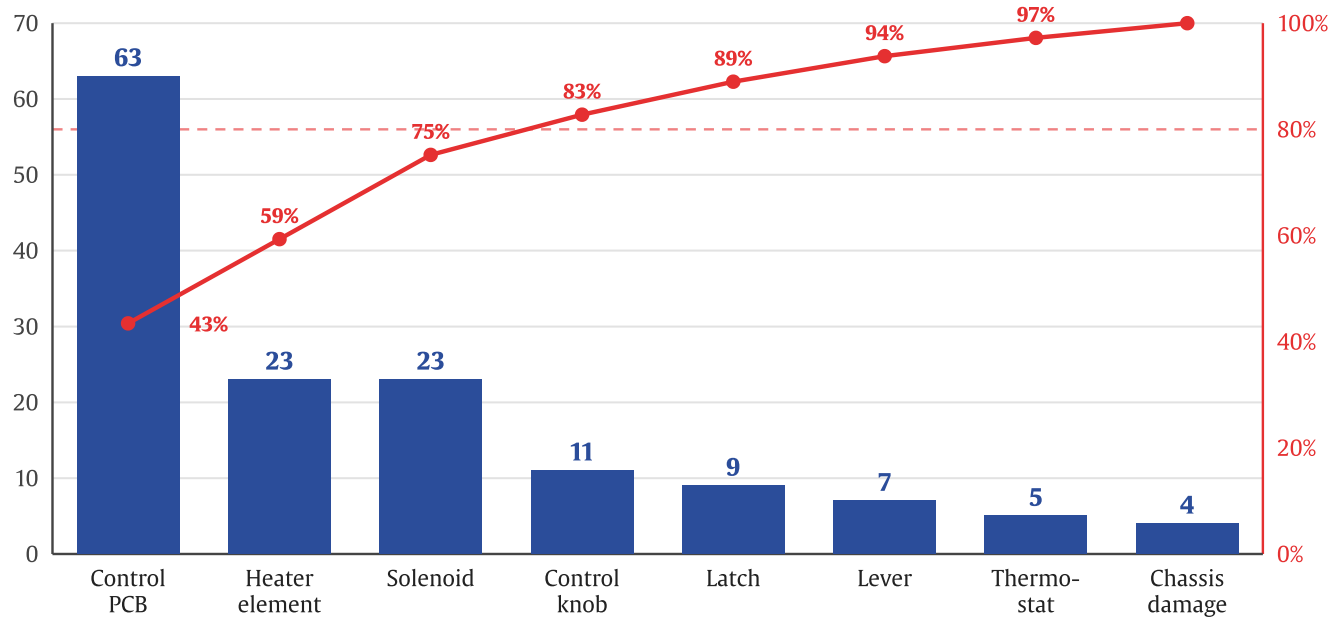


Two causes out of five explain **70 %** of the late arrivals. Leaving ten minutes earlier fixes more than any coffee strategy.

Your phone’s battery screen is a Pareto chart: sorted bars, and a few apps use most of the energy.

Pareto chart — the toaster line

The same 145 rejects, sorted



- **Control PCB alone: 43 %.**
- Top three — PCB, heater, solenoid — are **75 %** of all rejects, from 3 of 8 categories.
- Remove 6 in 10 PCB failures and the –36 a week target is met by that alone.

Three traps

Count ≠ cost. A scratch is cheap to rework, a dead PCB is a new board: draw the cost Pareto too.

A huge “Other” bar: badly chosen categories.
Redraw it after the fix: before / after is the proof.

Next question: why do the PCBs fail?

- **1** Every process varies. Quality management is largely the **management of variation**, and statistics is how variation is described.
- **2** The tools turn opinions into evidence, point at causes rather than symptoms, set priorities and separate signal from noise — and ISO 9001 §9.1 and §10.2 require exactly that.
- **3** **Flowchart:** draw what actually happens, not what the procedure says. The gap is itself a finding.
- **4** **Check sheet:** designed before collecting; who, when and where always recorded — or you cannot stratify later.
- **5** **Pareto chart:** sort, accumulate, attack the vital few first — and check that count and cost tell the same story.

DISCUSS

75 % of the toaster rejects come from three defect types. Would you put one team on the PCB, or three teams on the top three? What decides it?

PART 3

Find the causes

*Cause-and-effect diagram · Stratification · Scatter
diagram*

Cause-and-effect diagram

Tool 4 · What could be causing this effect? — Ishikawa, or “fishbone”

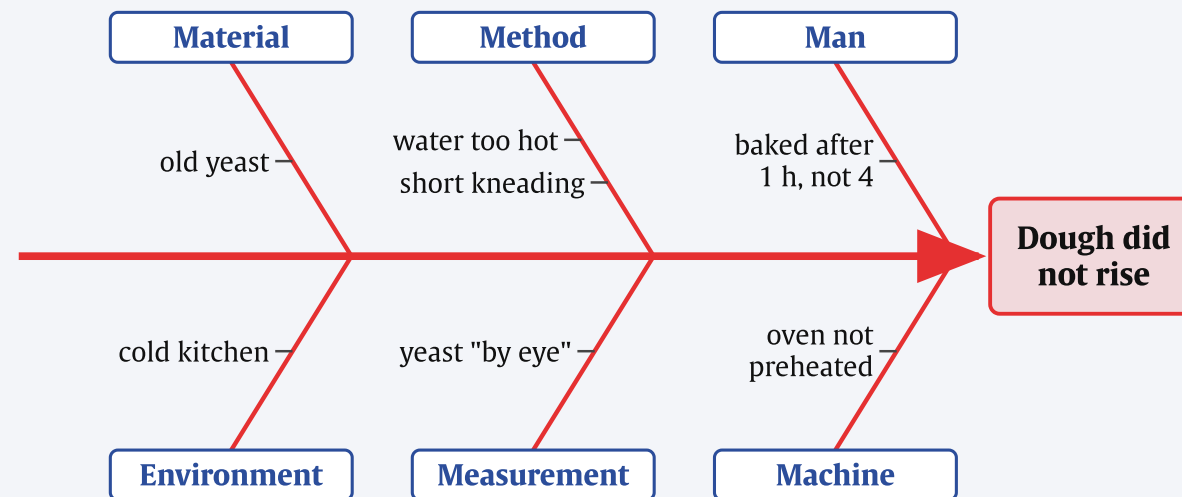
1. **The effect in the head** — specific and measurable: “PCB failures at final test”, not “bad quality”.
2. **The main bones** — categories of cause.
3. **Brainstorm as a team:** every idea on a bone. Ask “why?” again and again to grow smaller bones (the 5 Whys).
4. **Circle the likely causes** — then check them with data.

6M in manufacturing: Man, Machine, Method, Material, Measurement, Mother nature (environment). **8M** adds Management and Maintenance. Services often use People, Process, Policies, Place.

The diagram lists hypotheses, not facts

It tells you where to look. Stratification and scatter diagrams tell you whether you were right.

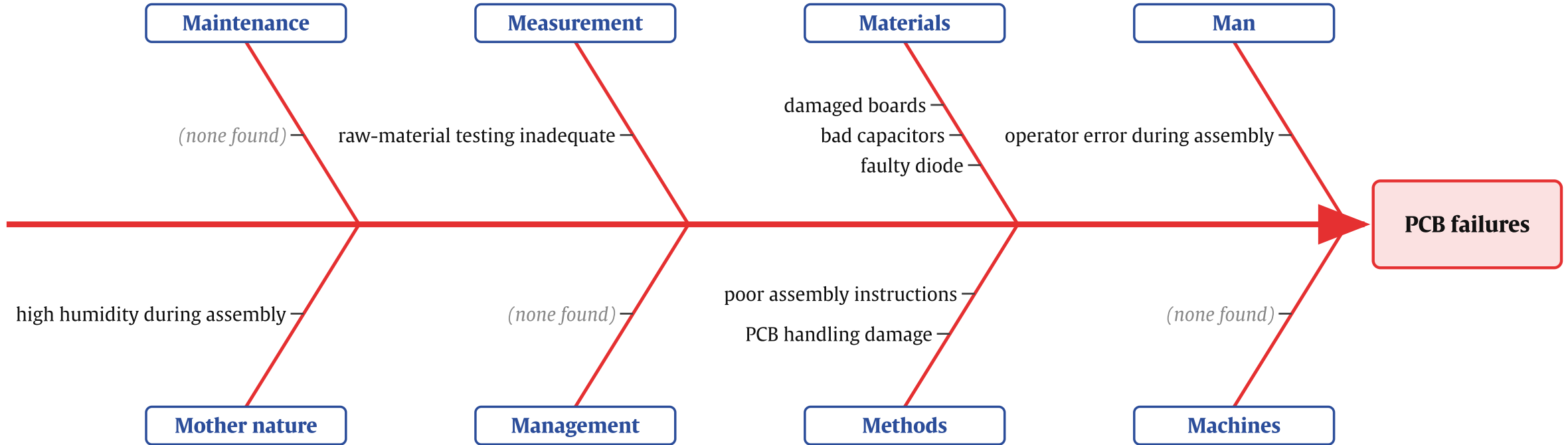
EVERYDAY · THE PIZZA DOUGH DID NOT RISE



5 Whys: why no rise? yeast dead → why? water too hot → why? used the kettle to save time → why? in a hurry → why? started too late. The root cause is the *plan*, not the yeast.

Cause and effect — PCB failures

The Pareto said “look at the PCB”. The team asks why. (8M)



Seven candidate causes. Which one first? Some can be checked against data we already have: the check sheet recorded the *day*, and the plant logs the *humidity*.

“None found” is a result too: the team looked at that bone and found nothing. An empty bone that nobody discussed is not the same thing.

Stratification

Tool 5 · Is it the same for every machine, shift, supplier, day?

Split the data by a factor that might matter, and compare the groups. An average over everything can hide the answer completely.

Typical strata: machine, line, mould cavity · shift, operator · supplier, material lot · day, time, season · temperature, humidity. It only works if the check sheet recorded who, when and where. Stratification is *paid for* at data collection.

EVERYDAY · YOUR COMMUTE

A month of commutes averages **34 min**, from 22 to 50. Split by mode: bus 40–50 min, bike 22–28 min. The “average commute” of 34 minutes never actually happens.

INDUSTRY · A TWO-CAVITY MOULD

The parts look widely spread. Split by cavity: each cavity is tight, but one is off-centre. Fix one cavity, not the whole process.

QUALITY INVESTIGATION · PCB FAILURES BY DAY

Day	PCB failures	Humidity	Stratum
Sun	6	18 %	dry mean 3.7 / day
Mon	4	15 %	
Tue	1	12 %	
Wed	10	60 %	humid mean 13.0 / day
Thu	17	91 %	
Fri	10	72 %	
Sat	15	85 %	

All **mean 9.0 / day — describes neither group**

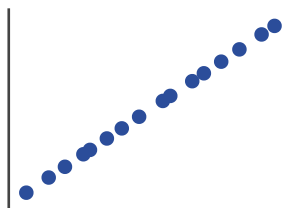
The strata are **3.5× apart**. “High humidity” from the fishbone is now the main suspect — but groups of 3 and 4 days are not proof yet.

Scatter diagram

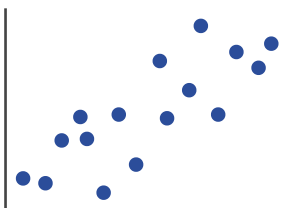
Tool 6 · Do two variables move together?

Plot pairs (x, y) measured on the same unit or day: x = a suspected cause, y = the effect. The strength of a *linear* relationship is the **Pearson correlation coefficient** r , from -1 to $+1$:

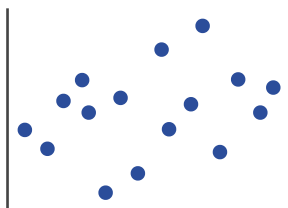
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2]}} \quad r^2 = \text{share of the variation of } y \text{ explained by a straight line in } x$$



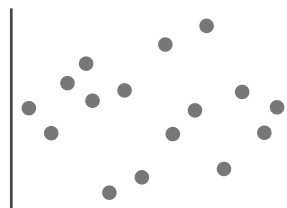
$r = +1.0$
perfect +



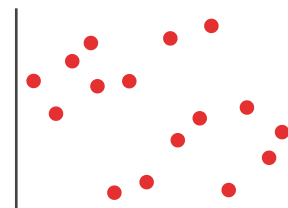
$r = +0.8$
strong +



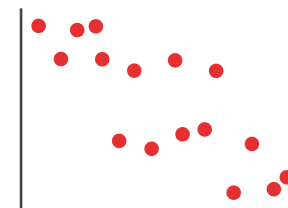
$r = +0.3$
weak +



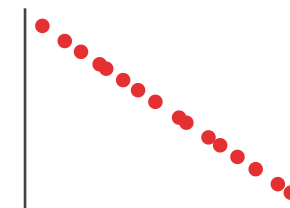
$r = 0$
none



$r = -0.3$
weak -



$r = -0.8$
strong -



$r = -1.0$
perfect -

EVERYDAY · HEATING AND THE WEATHER

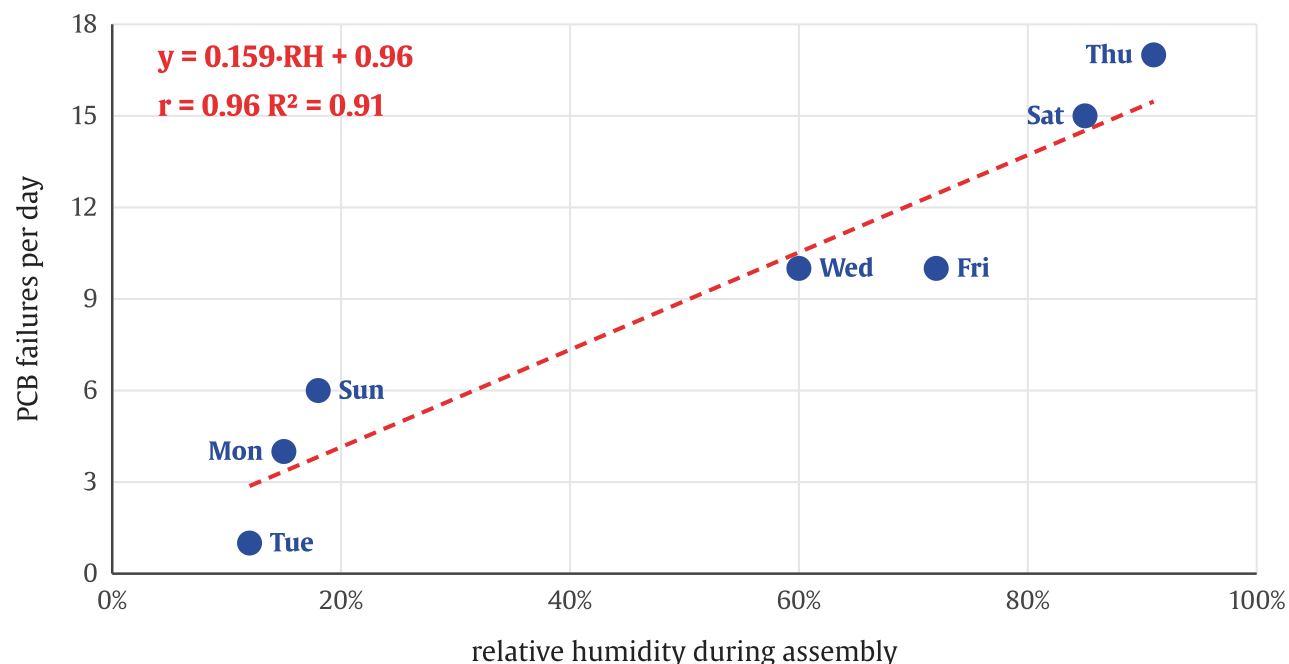
Outdoor temperature against the gas your flat burns each day: a strong **negative** r . Here the cause is real and obvious — colder days, more heating.

Correlation is not causation

Ice-cream sales and drowning accidents are positively correlated. Ice cream does not drown anyone: **summer** drives both. Always look for a hidden third variable. And $r \approx 0$ only means “no straight line” — there may still be a curve.

Scatter diagram — humidity vs PCB failures

Testing the fishbone's suspect with data we already had



- $r = 0.96$, $r^2 = 0.91$: humidity accounts for about 91 % of the day-to-day variation in PCB failures.
- Slope: every **+10 % humidity** $\approx$ **+1.6 failures** a day.
- A **mechanism** exists: boards and components absorb moisture before soldering — a known failure source for moisture-sensitive parts.

Still to do: change x and watch y

Control humidity in assembly, store boards dry, and see whether failures fall. That is the control chart in Part 5.

DISCUSS

Does the line mean that at 0 % humidity there would be about one failure a day? What is wrong with reading it there?

- **1 Cause and effect:** the effect in the head, causes on the bones (6M / 8M), 5 Whys to go deeper. It produces *hypotheses*, not answers.
- **2 Stratification:** split the data by machine, shift, supplier or condition. An overall average can describe no real group at all.
- **3 Scatter diagram:** suspected cause on x , effect on y . r measures the strength of a straight-line relation, r^2 the share explained.
- **4 Correlation is not causation:** look for a mechanism and a hidden third variable, and confirm by changing x .
- **5 The tools chain:** Pareto picks the problem, the fishbone lists candidates, stratification and scatter find the real one.

DISCUSS

A bank finds that branches with more staff receive more complaints. Should it cut staff? Which third variable would you check first?

PART 4

Describe variation

*Population and sample · Mean and standard deviation
· Histogram*

Population and sample

We almost never measure everything

1

Population

everything the process makes: all 10 000 bolts of the lot, size N

2

Sample

the n units you actually measure — chosen at random

3

Descriptive statistics

summarise the sample: mean, variance, standard deviation

4

Inferential statistics

from the sample, conclude something about the population

	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Variance	$\sigma^2 = \Sigma(x - \mu)^2 / N$	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$
Std. deviation	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Size	N	n

Greek letters for the population (usually unknown), Latin letters for the sample (computed).

EVERYDAY · TASTING THE SOUP

One spoonful tells you about the whole pot — **if you stir first**. Unstirred, a spoon from the top tells you about the top.

QUALITY INVESTIGATION · THE BOLT LOT

You measure 50 bolts out of 10 000. If all 50 come from the first hour of the morning shift, you have learned about the first hour, not the lot.

Mean, variance, standard deviation — by hand

Seven measurements: 104, 126, 102, 107, 121, 105, 119

x	$x - \bar{x}$	$(x - \bar{x})^2$
104	-8	64
126	14	196
102	-10	100
107	-5	25
121	9	81
105	-7	49
119	7	49
$\Sigma = 784$	0	564

$$\bar{x} = 784 / 7 = 112$$

$$s^2 = 564 / (7 - 1) = 94$$

$$s = \sqrt{94} = 9.69$$

Why divide by $n - 1$?

$\bar{x}$ is computed from the same seven numbers, so they sit closer to $\bar{x}$ than to the true μ . Dividing by n would underestimate the spread; $n - 1$ corrects it (Bessel's correction). With $n = 7$ it changes s by 8 %; with $n = 100$, by 0.5 %.

What s means

For a bell-shaped process, about **68 %** of values lie within $\bar{x} \pm s$, **95 %** within $\pm 2s$, **99.7 %** within $\pm 3s$. Here 112 ± 9.7 holds 5 of the 7 values.

Excel: =AVERAGE(), =STDEV.S() for a sample, =STDEV.P() for a whole population.

Histogram

Tool 7 · What does the variation look like?

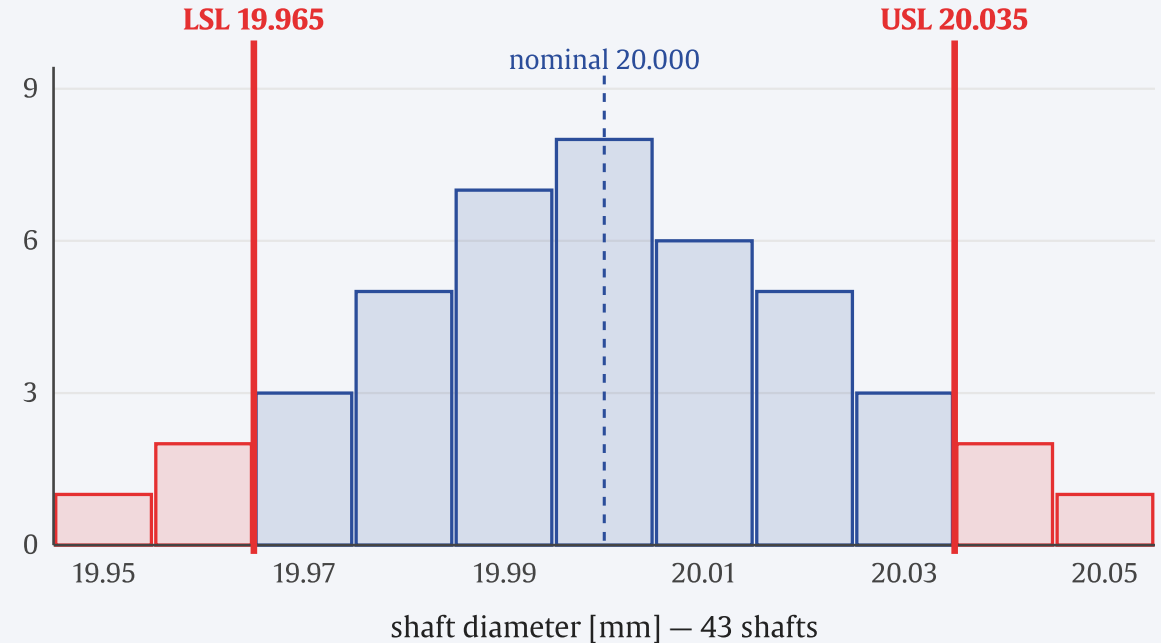
A bar chart of how often **continuous data** fall into equal-width classes. Every process varies in a **pattern**; the histogram makes the pattern visible, where a mean and an s alone would hide it.

- At least ~50 values; about $\sqrt{n}$ classes (5–20).
- Equal widths, no gaps between bars.
- Draw the **specification limits** on it: you see at once where you stand.

EVERYDAY · 40 COMMUTES

Most trips take 26–30 min, with a long tail out to 45. Traffic can only make you later, never much earlier: a **skewed** histogram, not a symmetric bell.

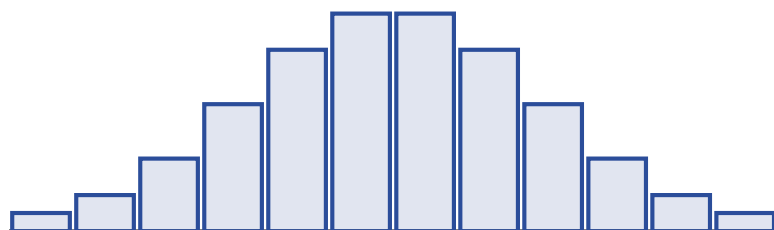
QUALITY INVESTIGATION · 43 TURNED SHAFTS



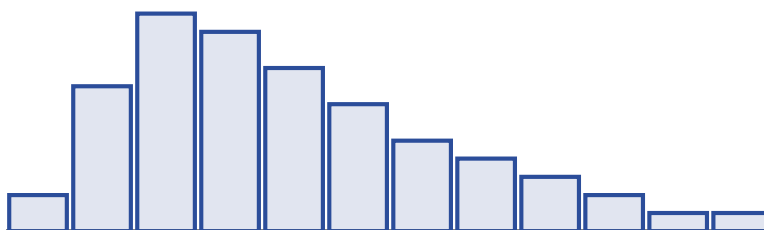
Specification $\varnothing 20 \pm 0.035$ mm. Centred on nominal, but **too wide**: both tails cross the limits, 6 of 43 shafts (14 %) out of specification. Re-centring will not help — the spread itself must shrink.

Reading a histogram

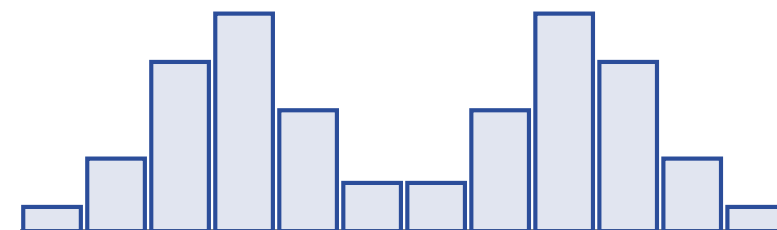
The shape tells you where to look



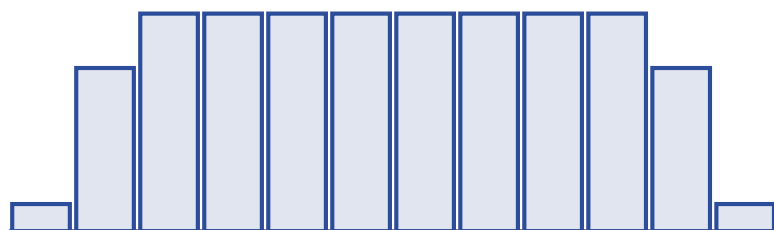
Bell (normal) — one stable process, many small independent causes. The reference case.



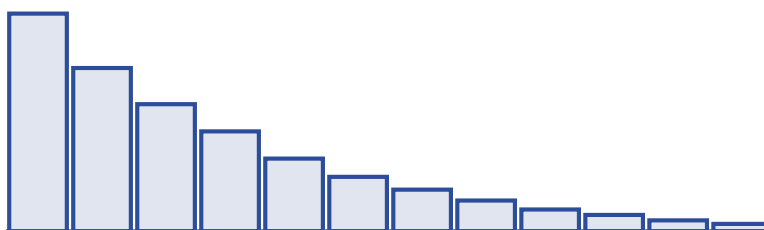
Skewed — a natural limit on one side (zero, a mechanical stop), or a tool wearing in one direction.



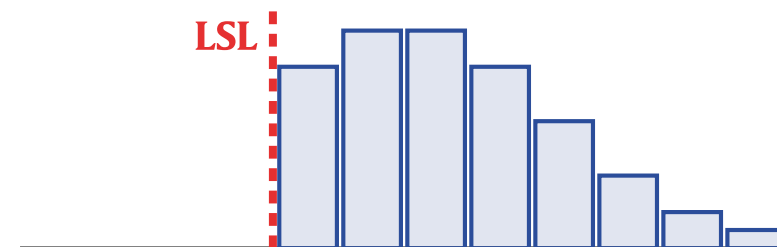
Two peaks — two populations mixed: two machines, shifts, suppliers. *Stratify*.



Plateau — several processes with different means blended, or a slow drift during the period.



Decaying — normal for times between failures or waiting times. Not every process should be a bell.



Cliff — one side cut off: out-of-spec parts were sorted out. The *process* is worse than the data you are shown.

DISCUSS

A supplier sends you a histogram of shaft diameters with a sharp cliff exactly at your lower limit, and no parts below. Good news or bad news?

PART 5

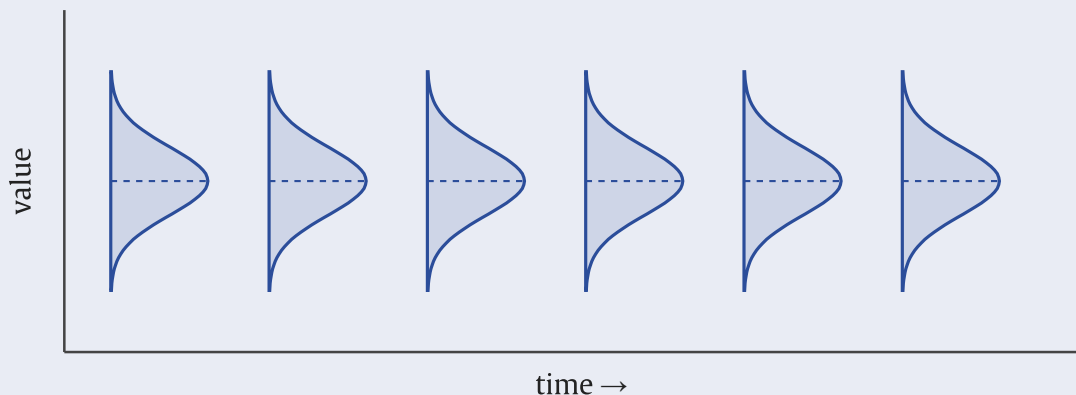
Control variation

Control charts · Process capability

Two kinds of variation

Tool 8 · Control chart — Shewhart's idea, 1924

Common cause — a stable process

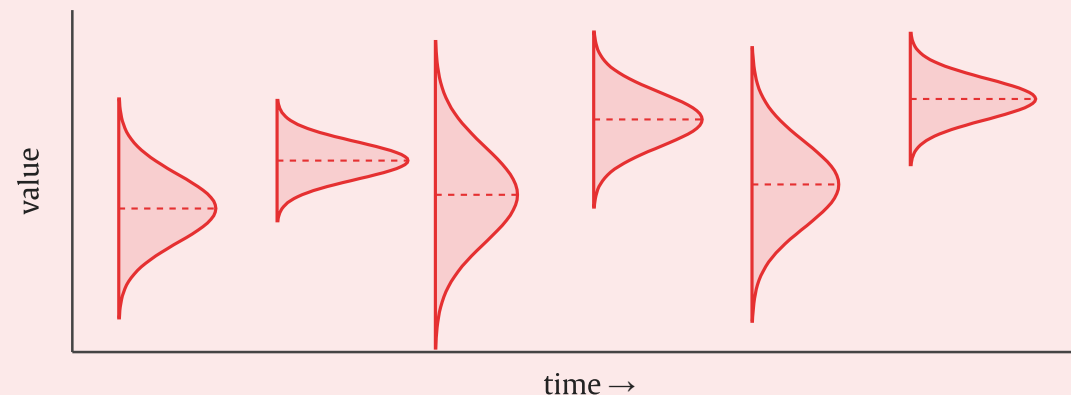


Many small causes, always present, built into the system. The output varies, but **predictably**. To reduce it you must **change the system** — a management decision.

EVERYDAY · THE BATHROOM SCALE

± 0.5 kg from day to day is water and breakfast: common cause. $+2$ kg after the holidays, and staying there, is a special cause.

Special cause — an unstable process



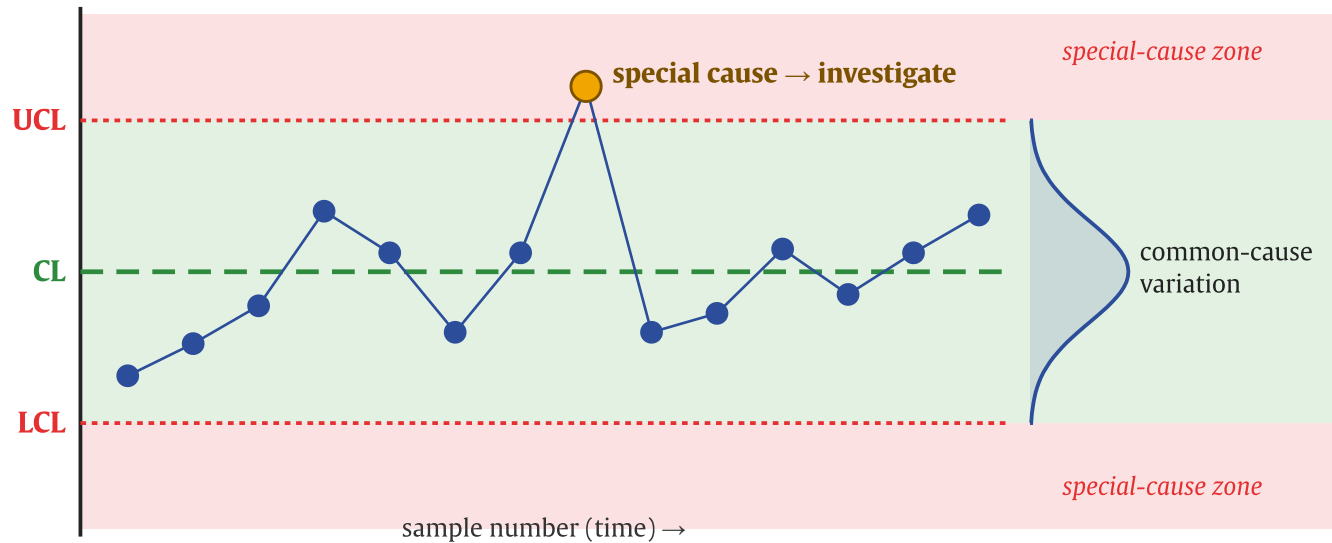
A specific, identifiable event: a worn tool, a new material lot, an untrained operator, a humid day. The output is **unpredictable**. Find the cause and **remove it**.

Two opposite mistakes

Treating common cause as special: adjusting the machine after every slightly high part. Deming's *funnel experiment* shows this **increases** variation. Or ignoring a real special cause. The control chart exists to avoid both.

Anatomy of a control chart

A centre line, and limits at $\pm 3\sigma$ of whatever is plotted



- **CL** — centre line: the process mean, from data.
- **UCL, LCL** — $CL \pm 3\sigma$. If the process is stable, 99.73 % of points fall inside; a point outside is very unlikely to be chance.

Signals of a special cause (Western Electric / Nelson rules):

- one point beyond a control limit
- 8 points in a row on one side of the CL — a shift
- 6 points in a row rising or falling — a drift
- 2 of 3 points beyond 2σ , same side

Control limits are not specification limits

Control limits are the **voice of the process**, calculated from data. Specification limits are the **voice of the customer**, written on the drawing. A process can be in control and still make scrap — or out of control and still within spec, for now.

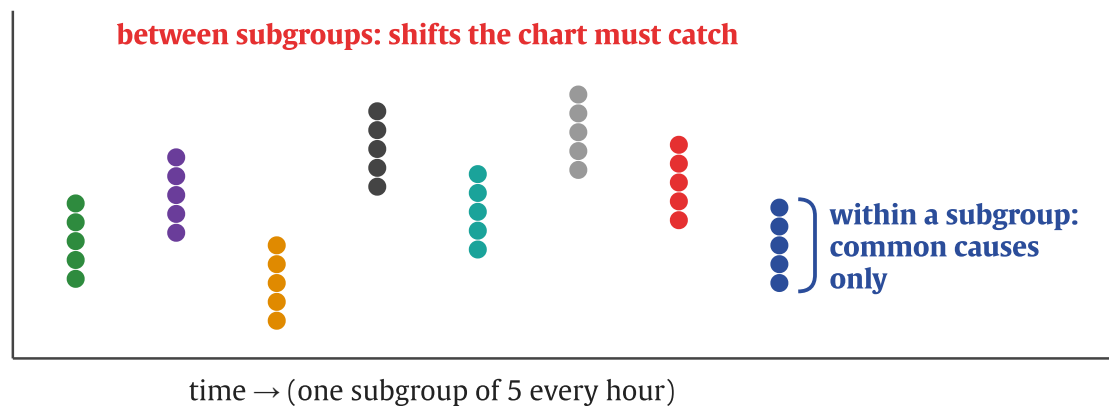
Which chart, and how to sample for it

The data type picks the chart; the subgroup sets its sensitivity

Data	Plotted	Chart
Measurements, subgroups of 2–9	mean and range	$\bar{X}-R$
Measurements, subgroups of 10+	mean and std. dev.	$\bar{X}-s$
Measurements, one at a time	value and moving range	I–MR
Defective units (pass / fail)	fraction or number defective	p / np
Defects (counts)	defects per unit or per period	u / c

Toaster line: number of PCB failures per day → **c chart**. Shifts measured in groups of 3 → $\bar{X}-R$.

Rational subgroup: a few units made under the same conditions — e.g. 3 to 5 consecutive parts, taken every hour.



The range *within* a subgroup then holds only common-cause variation. The limits are built from it, so they are tight enough to catch shifts *between* subgroups.

Wrong: one part from each of two machines in a subgroup. The machine difference inflates $\bar{R}$, the limits widen, and the chart goes blind.

$\bar{X}$ –R chart: the formulas

k subgroups of n units each

$\bar{X}$ chart — watches the process mean

$$CL = \bar{\bar{X}} = \Sigma \bar{X}_j / k \quad UCL = \bar{\bar{X}} + A_2 \bar{R} \quad LCL = \bar{\bar{X}} - A_2 \bar{R}$$

R chart — watches the spread within the process

$$CL = \bar{R} = \Sigma R_j / k \quad UCL = D_4 \bar{R} \quad LCL = D_3 \bar{R}$$

Estimate of the process σ

$$\sigma \approx \bar{R} / d_2$$

A_2, D_3, D_4, d_2 depend only on n . They turn the average range into $\pm 3\sigma$ limits, so nobody on the shop floor has to compute a standard deviation — the reason Shewhart used ranges in the age of pencils.

n	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.575	1.693
4	0.729	0	2.282	2.059
5	0.577	0	2.115	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078
15	0.223	0.347	1.653	3.472
20	0.180	0.415	1.585	3.735
25	0.153	0.459	1.541	3.931

$\bar{X}$ -R chart — worked example

Every hour, 3 consecutive parts are measured; ten hours of data

Hour	1	2	3	$\bar{X}$	R
1	7	4	4	5.0	3
2	6	3	8	5.7	5
3	9	10	12	10.3	3
4	7	8	4	6.3	4
5	11	6	10	9.0	5
6	7	4	7	6.0	3
7	10	8	11	9.7	3
8	10	11	10	10.3	1
9	5	11	9	8.3	6
10	8	4	6	6.0	4
Average				7.7	3.7

① **Each hour:** mean and range. Hour 1: $(7 + 4 + 4) / 3 = 5.0$; $R = 7 - 4 = 3$.

② **Centre lines:** $\bar{\bar{X}} = 7.7$ $\bar{\bar{R}} = 37 / 10 = 3.7$

③ **Constants for $n = 3$:** $A_2 = 1.023$, $D_3 = 0$, $D_4 = 2.575$, $d_2 = 1.693$

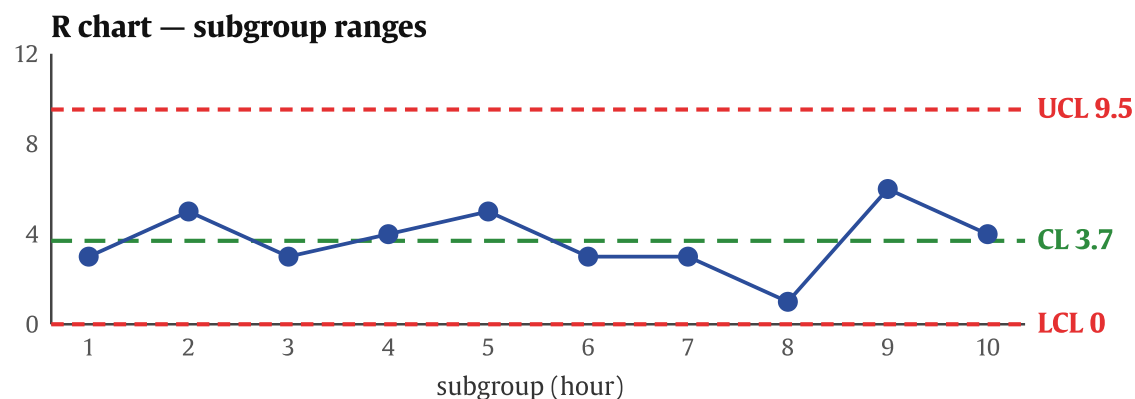
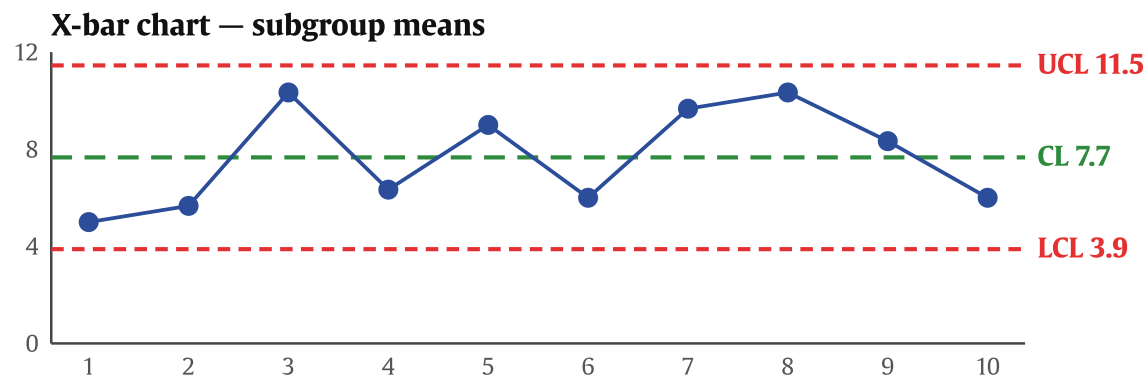
④ **$\bar{X}$ limits:** $7.7 \pm 1.023 \times 3.7 \rightarrow \text{UCL} = 11.5, \text{LCL} = 3.9$

⑤ **R limits:** $2.575 \times 3.7 \rightarrow \text{UCL} = 9.5$; $0 \times 3.7 \rightarrow \text{LCL} = 0$

⑥ **Process σ :** $3.7 / 1.693 \approx 2.2$ — we will need it for capability.

$\bar{X}$ -R chart — reading it

Same data, plotted against the limits we just computed



- **Read the R chart first.** The $\bar{X}$ limits are built from $\bar{R}$: if the spread is not stable, they mean nothing.
- All ranges below 9.5; all means between 3.9 and 11.5; no run of 8 on one side, no trend of 6.
- Verdict: **only common-cause variation** — the process is stable and predictable.

Stable does not mean good

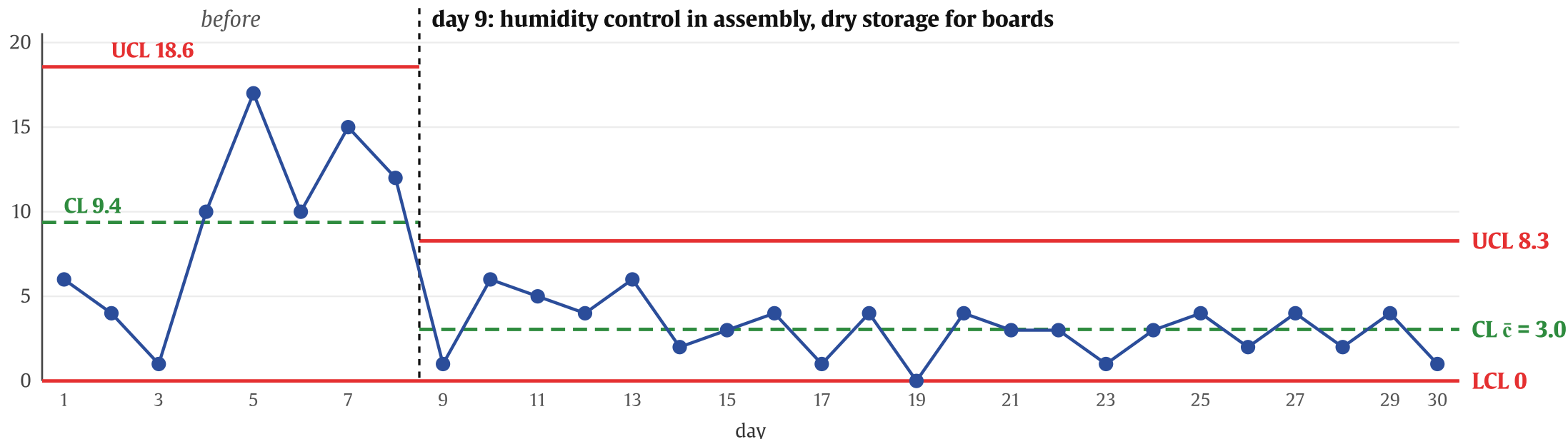
The chart says nothing about the customer. A perfectly stable process can make scrap every hour. That question is **capability**.

EVERYDAY · YOUR COMMUTE, CHARTED

One slow day is noise. Eight slow days in a row is a special cause — road works started — and a reason to change route.

Control chart — closing the toaster case

c chart of PCB failures per day, before and after the fix



c chart (defect counts)

$$CL = \bar{c}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL = \max(0, \bar{c} - 3\sqrt{\bar{c}})$$

For counts, the variance equals the mean.

Result

$\bar{c}$ from **9.4** to **3.0** failures a day: about **66** → **21** a week. One fix delivers more than the -36 target. New process, so **new limits**.

Act — hold the gain

Humidity limit written into the work instruction, dry cabinets for boards, and the chart **keeps running**.

Process capability: C_p and C_{pk}

Stable is not enough — can the process meet the specification?

Compare the **voice of the customer** (USL – LSL) with the **voice of the process** (6σ : $\pm 3\sigma$ holds 99.73 % of a normal process).

$$C_p = (USL - LSL) / 6\sigma$$

Potential: what the process could do if it were centred.

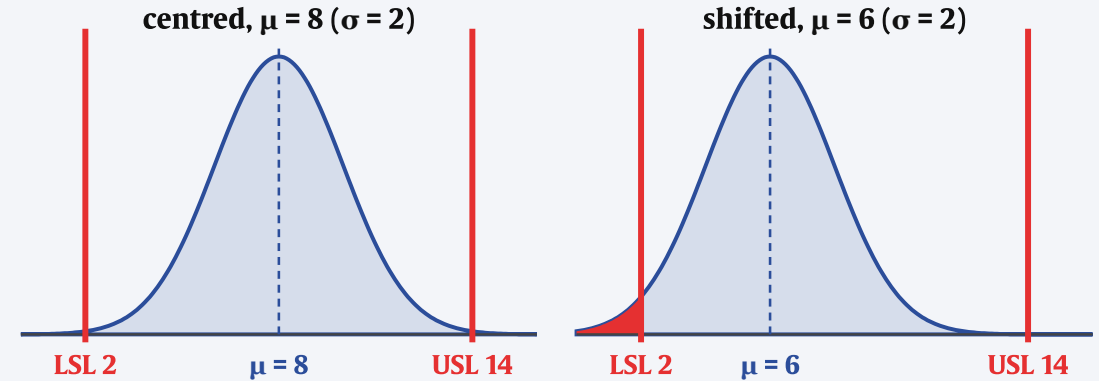
$$C_{pk} = \min[(USL - \mu) / 3\sigma, (\mu - LSL) / 3\sigma]$$

Actual: distance to the *nearest* limit. $C_{pk} \leq C_p$, equal only when centred.

EVERYDAY · PARKING IN YOUR GARAGE

The garage width is the specification; the car plus your wobble is the process spread. C_p : how many times the car fits the garage. C_{pk} : how close you are to the nearest wall. $C_p = 2$ with $C_{pk} = 0.3$ is a big garage — and a scratched mirror.

QUALITY INVESTIGATION · SAME SPREAD, DIFFERENT CENTRE



$C_p = C_{pk} = 1.00 \cdot 0.27\% \text{ out}$

$C_p = 1.00 \cdot C_{pk} = 0.67 \cdot 2.3\% \text{ out}$

$C_p = 12 / (6 \times 2) = 1.0$ in both. Shifted: $C_{pk} = \min(8/6, 4/6) = \mathbf{0.67}$.

Centred: $\min(6/6, 6/6) = \mathbf{1.0}$.

Same C_p , very different scrap. Centring a process is often far cheaper than reducing its σ .

Capability, performance, and what the numbers mean

Which σ you use changes the answer

- C_p, C_{pk} — σ within subgroups, $\sigma \approx \bar{R}/d_2$. Short term: what the process can do.
- P_p, P_{pk} — the same formulas with the overall s of all the data. Long term: what the customer actually got, drifts included.
- $P_{pk} \approx C_{pk}$ — stable over time. $P_{pk} \ll C_{pk}$ — the process moves between subgroups: look at the control chart.

Typical requirements

$C_{pk} \geq 1.33$ for an established process; ≥ 1.67 for new or safety-critical characteristics (common in automotive). Below 1: you are sorting, whether you admit it or not.

C_{pk}	nearest limit at	within spec	defects per million	about
0.33	1σ	68.27 %	317 311	1 in 3
0.67	2σ	95.45 %	45 500	1 in 22
1.00	3σ	99.73 %	2 700	1 in 370
1.33	4σ	99.994 %	63	1 in 16 000
1.67	5σ	99.99994 %	0.6	1 in 1.7 million
2.00	6σ	99.9999998 %	0.002	1 in 500 million

Centred, normally distributed process, both tails counted. Highlighted: our shifted example.

“Six Sigma = 3.4 defects per million”?

That figure assumes the mean drifts by 1.5σ in the long run, leaving 4.5σ to the nearest limit: it is $C_{pk} = 1.5$, not 2. Both numbers are “right” — know which convention you are reading.

- **1 Population and sample:** μ , σ are unknown; $\bar{x}$, s are computed. The sample must be representative — stir the soup.
- **2 Histogram:** the shape is information. Two peaks mean two processes; a cliff means somebody sorted.
- **3 Common vs special cause:** do not tamper with a stable process; hunt down special causes.
- **4 Control chart:** limits at $\pm 3\sigma$ come from the process, specification limits from the customer — never confuse them. Read R before $\bar{X}$.
- **5 Capability:** C_p is what you could do, C_{pk} what you do. 1.33 is the usual minimum; only meaningful for a *stable* process.

The toaster case, end to end

Seven tools, one story, no statistics beyond a spreadsheet

Step	Tool	What it told us
1	Flowchart	Rejects are found at final testing — collect data there.
2	Check sheet	145 rejects in a week, by defect type and by day.
3	Pareto chart	Control PCB failures are 43 % of rejects — start there.
4	Cause and effect	Seven candidate causes, humidity among them.
5	Stratification	Humid days: 13 failures a day. Dry days: 3.7.
6	Scatter diagram	$r = 0.96$ with humidity, and a physical mechanism to explain it.
—	The fix	Humidity control in assembly; dry storage for boards.
7	Control chart	Failures from 9.4 to 3.0 a day; new limits; chart kept running.

Each tool answered **one question** and handed the next tool its starting point. None needed more than a pencil or a spreadsheet. That is what Ishikawa meant by 95 %.

BACK TO THE START

At the beginning you wrote down what you would look at first. Would it have led you to the humidity?

Try it yourself

Three exercises to try at home

1 • Capability

The $\bar{X}$ -R process has mean 7.7 and $\sigma \approx 2.2$. The customer specifies 2 to 14.

Compute C_p and C_{pk} . Is the process capable? What would you change first: the mean or the spread?

2 • Your own Pareto

For one week, keep a check sheet of something in your life: what interrupted your study, why you were late, where your money went.

Record *when* as well as *what*. Draw the Pareto chart, then try one stratification.

3 • c chart

After the fix, the toaster line shows 3.0 PCB failures a day on average.

Compute UCL and LCL. Next Monday has 9 failures. Special cause or not? And if the next eight days are all at 4?

We go through the solutions together in class. Reference: *Juran's Quality Control Handbook*, chapters on basic statistical methods and statistical process control.

Questions?

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`github.com/LolloCappo/student-hub`